

NIER AGDM, AB, Canada

WMO: 714950

Lat: 51.3694N

Lon: 114.0992W

Elev: 1145

StdP: 88.31

Time Zone: -7.00 (NAM)

Period: 05-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-28.2	-24.8	-31.4	0.2	-27.9	-27.8	0.3	-24.6	13.1	-0.6	11.9	-3.4	3.2	310	0.731

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.4	27.5	15.9	25.6	15.3	23.8	14.6	18.0	24.1	16.8	23.1	15.7	21.9	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.7	12.8	21.4	14.3	11.7	19.9	13.1	10.8	18.3	55.6	24.2	51.6	23.3	48.0	22.0	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.2	10.6	9.3	-32.2	31.3	2.3	2.3	-33.8	32.9	-35.1	34.2	-36.4	35.5	-38.1	37.1
			-32.5	20.5	2.3	1.4	-34.2	21.6	-35.5	22.4	-36.7	23.2	-38.4	24.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.9	-5.5	-7.5	-2.1	3.2	9.2	12.6	15.5	14.5	10.5	4.5	-2.4	-7.1
	DBStd	10.19	8.08	8.53	7.45	5.18	4.36	3.02	2.84	3.37	4.91	5.22	7.45	8.02
	HDD10.0	2774	480	490	374	210	68	9	1	4	53	181	372	531
	HDD18.3	5301	738	723	633	454	283	172	96	124	237	429	621	789
	CDD10.0	533	0	0	1	6	44	88	171	145	68	10	1	0
	CDD18.3	18	0	0	0	0	1	1	8	6	2	0	0	0
	CDH23.3	484	0	0	0	2	23	31	174	177	74	3	0	0
	CDH26.7	89	0	0	0	0	1	3	31	42	12	0	0	0
(14) Wind	WSAvg	4.4	5.1	4.3	4.4	4.8	4.8	4.4	3.7	3.5	3.9	4.4	4.7	4.7
(15) Precipitation	PrecAvg	472	14	11	19	31	69	101	69	63	43	20	17	15
	PrecMax	636	27	26	54	75	127	298	156	117	107	43	52	38
	PrecMin	318	1	0	6	9	25	28	23	13	6	4	3	3
	PrecStd	67	7	7	10	18	29	53	36	27	24	11	11	9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.7	12.2	15.2	21.3	26.0	26.5	29.5	30.2	28.3	21.5	15.7	10.8
		MCWB	4.0	4.5	6.0	9.0	11.6	16.4	17.2	15.7	13.7	10.2	6.5	3.1
	2%	DB	8.0	8.2	12.4	18.1	23.2	23.8	27.3	27.5	25.6	18.0	11.9	7.4
		MCWB	2.7	2.4	4.8	8.0	11.3	14.5	17.2	15.7	13.2	8.9	5.2	2.3
	5%	DB	6.4	5.8	9.8	15.0	20.8	22.0	25.2	25.2	22.9	15.1	9.4	5.5
		MCWB	1.7	1.2	3.3	6.2	10.3	13.3	16.9	15.1	12.5	7.4	3.7	1.2
	10%	DB	5.0	3.6	7.6	12.0	18.5	20.0	23.3	22.9	19.9	12.8	7.0	3.6
		MCWB	0.9	-0.3	2.3	4.7	9.2	12.5	16.0	14.8	11.5	6.2	2.2	-0.3
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.4	4.8	6.9	9.7	13.2	18.0	20.5	18.6	15.0	11.2	7.6	4.2
		MCDB	9.5	11.4	13.4	20.4	22.5	23.6	26.1	25.1	24.6	19.8	15.0	8.9
	2%	WB	2.9	2.7	5.2	8.2	11.7	15.8	18.7	17.2	14.0	9.5	5.5	2.6
		MCDB	7.6	7.8	11.4	17.5	21.2	21.9	24.9	24.0	23.1	16.7	11.4	7.0
	5%	WB	2.0	1.2	3.7	6.6	10.7	14.4	17.5	16.0	12.9	8.0	4.1	1.3
		MCDB	6.1	5.5	9.0	14.1	19.4	20.1	23.4	23.2	21.3	14.5	9.0	5.4
	10%	WB	0.9	-0.2	2.4	5.0	9.7	13.3	16.4	14.9	11.7	6.6	2.5	-0.2
		MCDB	4.8	3.2	7.3	10.9	17.4	18.6	22.2	21.8	19.4	12.1	6.6	3.3
(35) Mean Daily Temperature Range	5% DB	MDBR	10.3	10.6	10.5	11.5	12.5	11.5	13.4	13.5	12.9	11.7	10.3	9.6
		MCDBR	10.4	12.6	13.9	16.7	16.5	15.4	16.7	17.8	18.0	14.7	12.8	10.5
		MCWBR	6.7	8.3	8.1	8.6	7.5	7.7	8.7	8.2	7.9	7.9	7.4	7.0
	5% WB	MCDBR	10.4	12.3	13.1	15.6	14.9	13.3	14.9	15.8	16.4	14.1	12.6	10.4
		MCWBR	6.9	8.2	8.0	8.3	7.1	7.3	8.6	8.2	7.6	7.7	7.7	7.2
(40) Clear-Sky Solar Irradiance	taub		0.248	0.263	0.276	0.328	0.336	0.341	0.337	0.351	0.307	0.275	0.249	0.234
	taud		2.410	2.445	2.483	2.346	2.385	2.409	2.419	2.390	2.479	2.553	2.498	2.395
	Ebn at Noon		802	888	938	912	918	913	910	878	888	864	803	762
	Edh at Noon		57	73	86	111	112	111	108	106	86	65	52	49
(44) All-Sky Solar Radiation	RadAvg		1.12	2.10	3.40	4.63	5.49	5.83	6.38	5.23	3.69	2.27	1.22	0.87
	RadStd		0.07	0.16	0.24	0.34	0.42	0.42	0.39	0.37	0.41	0.24	0.09	0.07

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (62 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page